Slow and geometric terminal rates for scalar AdGD2

Abstract

The terminal behavior of the original scalar AdGD2 rule depends sharply on the structure of the minimum. For every prescribed positive sequence $r_k \to 0$, one fixed convex C^∞ objective with derivative Lipschitz constant exactly one and unique minimizer zero has a convergent trajectory, initialized by $x_0 = \alpha_0 = 1$, whose terminal gradient is neither $O(r_k)$ nor $O(1/k)$. The same construction gives arbitrarily slow terminal objective gaps, while preserving weighted gradient-square summability. In contrast, for a finite convex C^1 objective with locally Lipschitz derivative $g(x) \sim ax^p$ at its unique minimizer, where $a > 0$ and $p \geq 3$ is odd, every positive initial step gives either finite termination or exact geometric terminal equivalents. We prove from the actual adaptive minimum that the curvature candidate is eventually infinite, the sign is eventually constant, and $\alpha_k \sim Aq^k$ with $A > 0$ and $q = (1 + \sqrt{11/3})/2$. The iterate, gradient and objective equivalents have respective bases $q^{-1/(p-1)}$, $q^{-p/(p-1)}$ and $q^{-(p+1)/(p-1)}$. The finite-order result needs only a derivative-value asymptotic; it does not assume a second-derivative asymptotic.

Keywords: adaptive gradient descent; terminal rates; smooth convex optimization; finite-order minima

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1 Terminal rates for the original scalar rule

Let $f : \mathbb{R} \to \mathbb{R}$ be convex and differentiable, and write $g = f'$. Algorithm 2 of the final NeurIPS 2024 paper of Malitsky–Mishchenko [1], called AdGD2, takes the initialization

$$x_0 \neq 0, \qquad \alpha_0 > 0, \qquad \theta_0 = \tfrac{1}{3}, \qquad x_1 = x_0 - \alpha_0 g(x_0), \tag{1}$$

and, for $k \geq 1$, the updates

$$\begin{aligned} L_k &= \frac{|g(x_k) - g(x_{k-1})|}{|x_k - x_{k-1}|}, \\ \alpha_k &= \min\left\{ \alpha_{k-1}\sqrt{\tfrac{2}{3} + \theta_{k-1}},\ \frac{\alpha_{k-1}}{\sqrt{[2\alpha_{k-1}^2 L_k^2 - 1]_+}} \right\}, \\ x_{k+1} &= x_k - \alpha_k g(x_k), \qquad \theta_k = \alpha_k/\alpha_{k-1}. \end{aligned} \tag{2}$$

Here $[t]_+ = \max\{t, 0\}$ and, as in the source, a positive number divided by zero is $+\infty$. If an iterate has zero gradient, the run terminates there; no step or secant is requested afterward. In the unique-minimizer setting below, this is exactly termination at zero. The infinite trajectories we study have nonzero secant denominators. The measured curvature L_k is not replaced by a global Lipschitz bound.

Section 5 of [1] identifies the $O(1/k)$ last-gradient norm known for fixed-step gradient descent as an open rate question for the adaptive method. Its Theorem 2 proves iterate convergence and a best-objective bound, and its proof yields weighted gradient-square summability. The first theorem answers the $O(1/k)$ benchmark negatively, with fixed-function quantifiers and infinite differentiability. The second gives exact terminal asymptotics under a finite-order hypothesis. These conclusions concern different classes of minima and are compatible.

Theorem 1. *For every sequence $(r_k)_{k\geq 1}$ of positive real numbers tending to zero, there exists one fixed convex C^∞ function $f : \mathbb{R} \to \mathbb{R}$ whose derivative has global Lipschitz constant exactly one and whose unique minimizer is zero, such that the actual iteration (1)–(2), with $x_0 = \alpha_0 = 1$, is defined for every index, never reaches zero gradient, and satisfies*

$$\begin{gathered} x_k \longrightarrow 0, \qquad g(x_k) \longrightarrow 0, \qquad \sum_{k=0}^{\infty} \alpha_k |g(x_k)|^2 < \infty, \\ \limsup_{k\to\infty} k|g(x_k)| = \infty, \qquad \limsup_{k\to\infty} \frac{|g(x_k)|}{r_k} = \infty. \end{gathered} \tag{3}$$

The function may depend on the prescribed sequence r, but remains the same throughout its infinite run.

The quantifier order is $\forall r\ \exists f_r$. The theorem excludes a common prescribed iteration-index rate even if its multiplicative constant and starting index depend on the entire instance. It does not say that one function defeats all null sequences, or that an individual convergent trajectory has no rate. The divergence in (3) occurs along a subsequence.

Theorem 2. *Let $f : \mathbb{R} \to \mathbb{R}$ be finite, convex and C^1, with locally Lipschitz derivative $g = f'$, and let zero be its unique minimizer. Suppose that, for some $a > 0$ and an odd integer $p \geq 3$,*

$$\lim_{x\to 0,\ x\neq 0} \frac{g(x)}{ax^p} = 1. \tag{4}$$

For every $x_0 \neq 0$ and $\alpha_0 > 0$, use exactly (1)–(2) and its termination convention. The run either terminates at zero in finitely many steps or is well-defined and nonzero at every index, with $x_k \to 0$. In the infinite case, there is a finite K such that, for all $k \geq K$,

$$2\alpha_{k-1}^2 L_k^2 < \frac{25}{32} < 1, \qquad \alpha_k = \alpha_{k-1}\sqrt{\tfrac{2}{3} + \theta_{k-1}}. \tag{5}$$

Thus the actual curvature candidate is permanently $+\infty$, and the sign of x_k is eventually a constant $\sigma \in \{-1, 1\}$. Put

$$q = \frac{1 + \sqrt{11/3}}{2}, \qquad \zeta_p = 1 - q^{-1/(p-1)}. \tag{6}$$

There is an instance-dependent $A \in (0,\infty)$, with the original global iteration index, such that

$$\theta_k \longrightarrow q, \qquad \frac{\alpha_k}{q^k} \longrightarrow A, \qquad \frac{\alpha_k g(x_k)}{x_k} \longrightarrow \zeta_p. \tag{7}$$

With $D = (\zeta_p/(aA))^{1/(p-1)} \in (0,\infty)$, the same trajectory satisfies

$$\begin{aligned} q^{k/(p-1)} x_k &\longrightarrow \sigma D, \\ q^{pk/(p-1)} g(x_k) &\longrightarrow \sigma a D^p, \\ q^{(p+1)k/(p-1)} (f(x_k) - f(0)) &\longrightarrow \frac{aD^{p+1}}{p+1}. \end{aligned} \tag{8}$$

In particular, the consecutive ratios satisfy

$$\begin{aligned} \frac{|x_{k+1}|}{|x_k|} &\longrightarrow q^{-1/(p-1)}, \qquad \frac{|g(x_{k+1})|}{|g(x_k)|} \longrightarrow q^{-p/(p-1)}, \\ \frac{f(x_{k+1}) - f(0)}{f(x_k) - f(0)} &\longrightarrow q^{-(p+1)/(p-1)}. \end{aligned} \tag{9}$$

Theorem 2 requires neither global Lipschitz continuity of g, analyticity, a separate asymptotic for f'', nor global strong convexity. It includes arbitrary initial points and arbitrary positive initial steps. Its positive amplitude and equivalents refer to one infinite trajectory; they are not uniform finite-horizon complexity estimates. We make no claim for $p = 1$, for multidimensional objectives, or for floating-point implementations.

The closest inspected same-rule extension [2] explicitly includes AdGD2 and retains qualitative stationarity and weighted square summability. Later last-objective and improved best-gradient bounds in [3, 4] concern different adaptive rules, using function-value curvature and different growth controls. The source's local-curvature examples without a growth cap [1, Section 2] also differ: both printed caps are retained here. Fixed-function slow convergence is an established theme; for example, [5] studies energy decay for gradient flows and fixed-step gradient descent under different observables and rate constraints. The contribution here is the compatible infinite realization for the exact AdGD2 terminal gradient, including smoothness at the accumulating intervals and their minimizer.

2 Smooth scalar functions and a growth sequence

Define

$$\beta(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0, \end{cases} \qquad \chi(t) = \frac{\beta(t)}{\beta(t) + \beta(1-t)}, \qquad \psi(a) = \int_0^a e^{-1/u^2}\, du \quad (a \ge 0), \tag{10}$$

where the integrand at zero is zero. The function β is smooth and flat at zero: each positive-side derivative is a polynomial in $1/t$ times $e^{-1/t}$ and tends to zero faster than any power of t. The denominator defining χ is everywhere positive. Thus χ is smooth,

equals zero on $(-\infty, 0]$ and one on $[1, \infty)$, and all its positive-order derivatives vanish at both endpoints. On $(0, 1)$,

$$\chi'(t) = \chi(t)(1 - \chi(t))\big(t^{-2} + (1-t)^{-2}\big) > 0.$$

Moreover $\chi(t) + \chi(1-t) = 1$, so $\int_0^1 \chi(t)\, dt = 1/2$. Define the finite constants

$$C_j = \max_{0 \le t \le 1} |\chi^{(j)}(t)| \quad (j \ge 1), \qquad K_0 = 1, \qquad K_n = \max\{1, C_1, \dots, C_n\}. \tag{11}$$

The same exponential derivative argument makes ψ smooth and flat at zero. For $a > 0$,

$$0 < \psi(a) \le a e^{-1/a^2} < a, \qquad 0 < \psi'(a) = e^{-1/a^2} < 1. \tag{12}$$

Given $A, v > 0$, define auxiliary numbers

$$\begin{gathered} A_0 = A, \quad v_0 = v, \quad v_{j+1} = \sqrt{\tfrac{2}{3} + v_j}, \quad A_{j+1} = A_j v_{j+1}, \\ S_0 = 0, \qquad S_N = \sum_{j=0}^{N-1} A_j \quad (N \ge 1). \end{gathered} \tag{13}$$

They will specify the objective; their agreement with the actual adaptive steps will be proved later.

Lemma 1. *For every positive A, v, with $q = (1 + \sqrt{11/3})/2$, the sequence* (13) *satisfies $v_j \to q$, $S_N \to \infty$, and $S_{N-1}/S_N \to 1/q$.*

Proof. The map $t \mapsto \sqrt{2/3 + t}$ moves each positive t monotonically toward its unique positive fixed point $q > 1$. Thus $A_j/A_{j-1} = v_j \to q > 1$, and A_j, S_N diverge. Also

$$\frac{S_{N-1}}{S_N - A_0} = \frac{\sum_{j=0}^{N-2} A_{j+1}(A_j/A_{j+1})}{\sum_{j=0}^{N-2} A_{j+1}} \longrightarrow \frac{1}{q}.$$

Discard a fixed finite prefix and bound the remaining ratios uniformly near $1/q$ to justify this weighted limit. Since $A_0/S_N \to 0$, the stated ratio follows. ∎

Fix $c = 1/8$. For a growth sequence and $b > 0$, define, for integers $N \ge 2$,

$$\begin{gathered} e(N) = \frac{9b}{8S_N}, \qquad p(N) = b - e(N)S_{N-1}, \qquad a = cb, \qquad h = \psi(a), \\ D(N) = \sqrt{2(h + e(N))^2 - e(N)^2}, \qquad B(N) = \frac{p(N) + a}{D(N)}, \\ b^+(N) = B(N)h - a, \qquad d(N) = \frac{e(N)}{D(N)}. \end{gathered} \tag{14}$$

For a stage index $n \ge 0$, call N admissible if

$$\begin{gathered} \frac{b}{5} < p(N) < \frac{b}{4}, \qquad \frac{b}{10} < b^+(N) < \frac{b}{6}, \qquad 0 < d(N) < \frac{1}{4}, \\ 0 < e(N) \le \frac{b}{100}, \qquad e(N) \le \frac{b}{60(1 + C_1)}, \qquad e(N) \le \frac{b^{n+2}}{30^n(1 + K_n)}. \end{gathered} \tag{15}$$

Lemma 2. *For each fixed positive A, v, b and each fixed stage index n, every sufficiently large integer N is admissible.*

Proof. Lemma 1 gives $e(N) \to 0$ and

$$\frac{p(N)}{b} \to P = 1 - \frac{9}{8q}, \qquad \frac{b^+(N)}{b} \to Z = \frac{P + 1/8}{\sqrt{2}} - \frac{1}{8}, \qquad d(N) \to 0.$$

Here $h > 0$ is fixed, and cancels in the limit of $B(N)h$. Evaluating $t^2 - t - 2/3$ gives $45/32 < q < 3/2$, hence $1/5 < P < 1/4$. Consequently

$$\frac{13}{40\sqrt{2}} - \frac{1}{8} < Z < \frac{3}{8\sqrt{2}} - \frac{1}{8}.$$

The lower endpoint exceeds $1/10$ since $13 > 9\sqrt{2}$; the upper endpoint is below $1/6$ since $9 < 7\sqrt{2}$. The remaining bounds on $e(N)$ are fixed positive thresholds. Finally $D(N)^2 = 2h^2 + 4he(N) + e(N)^2 > 0$. ∎

3 Definition of one fixed objective

Fix the positive null sequence r in Theorem 1. Set $b_0 = 1/2$. For stage zero, use (13) with $(A, v) = (1, 1)$, and let $N_0 \geq 2$ be the least admissible integer for $n = 0$. Give the resulting quantities in (14) the subscripts zero, writing $b_1 = b^+(N_0)$ and $\delta_0 = d(N_0)$. Denote its growth terms and sums by $A_{0,j}, v_{0,j}, S_{0,j}$. Set

$$t_0 = 1, \qquad t_1 = N_0 + 2.$$

In particular, the seed sequence begins at actual time one.

Suppose all data through stage $n - 1$, and b_n, t_n, are fixed, where $n \geq 1$. Define

$$\tau_n = \sqrt{\tfrac{2}{3} + \delta_{n-1}}, \qquad \Gamma_n = h_{n-1}\left[\sqrt{\frac{1 + \tau_n^{-2}}{2}} - 1\right]. \tag{16}$$

Use (13) with $(A, v) = (B_{n-1}\tau_n, \tau_n)$, and denote its terms by $A_{n,j}, v_{n,j}, S_{n,j}$. Choose N_n as the least admissible integer for stage n that also satisfies

$$\begin{aligned} e(N_n) &\leq \frac{e_{n-1}}{2}, \qquad e(N_n) \leq \frac{\Gamma_n}{2}, \\ N_n\psi(cb_n) &\geq n, \qquad r_{t_n+N_n} \leq \frac{\psi(cb_n)}{n}. \end{aligned} \tag{17}$$

Assign the corresponding values the subscripts n, so $a_n = cb_n$, $h_n = \psi(a_n)$, $b_{n+1} = b^+(N_n)$ and $\delta_n = d(N_n)$, and set $t_{n+1} = t_n + N_n + 1$.

Every least integer exists. Lemma 2 supplies the seed. At a subsequent stage, all preceding numbers are finite and positive, and

$$\sqrt{2/3} < \tau_n < \sqrt{11/12} < 1,$$

so $\Gamma_n > 0$. Every sufficiently large N is admissible, $e(N) \to 0$, and the fixed number $\psi(cb_n)$ is positive. All four extra conditions in (17) therefore hold eventually; for the last, t_n is already fixed and $r_k \to 0$. Monotonicity of r is unnecessary. Induction defines the whole recursion before the algorithm is run. In particular,

$$\frac{b_n}{10} < b_{n+1} < \frac{b_n}{6} < \frac{b_n}{5} < p_n < \frac{b_n}{4} < b_n, \qquad 0 < e_{n+1} \le \frac{e_n}{2}. \tag{18}$$

Put $\ell_n = p_n - b_{n+1} > 0$. Define g on the whole real line by

$$\begin{aligned} g(x) &= -\psi(-x) && (x \le 0), \\ g(x) &= e_0 + \int_{b_0}^{x} \chi\left(\frac{u - b_0}{2e_0}\right) du && (x \ge b_0), \\ g(x) &= e_n && (p_n \le x \le b_n), \\ g(x) &= e_{n+1} + (e_n - e_{n+1})\chi\left(\frac{x - b_{n+1}}{\ell_n}\right) && (b_{n+1} \le x \le p_n), \end{aligned} \tag{19}$$

where the last two lines apply for every $n \ge 0$. Finally let

$$f(x) = \int_0^x g(u)\, du. \tag{20}$$

The endpoint values agree. Since $b_n \le b_0 6^{-n} \to 0$, the intervals $[b_{n+1}, b_n]$ cover $(0, b_0]$.

4 Global smoothness and normalization

The cutoff is flat at both endpoints, so every positive-order derivative of each connector in (19) agrees with the adjacent plateau. There is no positive accumulation point of these junctions. Thus g is smooth on $(0, b_0]$, positive and nondecreasing there. Equation (18) gives

$$\ell_n = p_n - b_{n+1} > \frac{b_n}{30}. \tag{21}$$

On its nth connector, for each $j \ge 1$,

$$|g^{(j)}(x)| \le e_n C_j \left(\frac{30}{b_n}\right)^j. \tag{22}$$

In particular the first smallness bound in (15) involving C_1 gives

$$0 \le g'(x) \le \frac{30 e_n C_1}{b_n} \le \frac{C_1}{2(1 + C_1)} < \frac{1}{2}.$$

For a fixed $j \ge 1$ and $n \ge j$, the last bound in (15) yields

$$|g^{(j)}(x)| \le \frac{C_j}{1 + K_n}\, 30^{j-n} b_n^{n+2-j} \le b_n^2. \tag{23}$$

The same estimate holds on the plateau, where the derivative is zero. Also $0 < g(x) \le e_n \le b_n^2$ throughout $[b_{n+1}, b_n]$. Since $x \ge b_{n+1} > b_n/10$ there, for every fixed $j \ge 0$,

$$\frac{|g^{(j)}(x)|}{x} \le 10 b_n \longrightarrow 0 \quad \text{as } x \downarrow 0, \tag{24}$$

using the separate value estimate when $j = 0$.

For completeness, these estimates prove actual smooth extension at zero, rather than only matching formal jets. Let G_j be the ordinary jth derivative on $x > 0$, with $G_j(0) = 0$. Equations (23)–(24) show continuity at zero and

$$\lim_{x \downarrow 0} \frac{G_j(x) - G_j(0)}{x} = 0 = G_{j+1}(0).$$

Induction identifies every G_j as the corresponding derivative of the extension. The negative branch $-\psi(-x)$ has exactly the same zero jet, by (10). Hence g is C^∞ across zero.

The upper branch also joins the plateau smoothly at b_0. Its derivative is $\chi((x - b_0)/(2e_0)) \in [0, 1]$. The identity $\int_0^1 \chi = 1/2$ gives

$$g(x) = x - b_0 \qquad (x \ge b_0 + 2e_0). \tag{25}$$

Together with (12) and the connector estimates, this proves $0 \le g' \le 1$ globally. The mean value theorem gives the global Lipschitz bound one; (25) makes its least possible constant exactly one. Consequently f in (20) is finite, convex and C^∞. Since $g < 0$ on $(-\infty, 0)$ and $g > 0$ on $(0, \infty)$, its unique minimizer is zero and $f(0) = 0$.

5 Verification of the actual infinite trajectory

We now verify every branch of (2) on this fixed function. The key distinction is between the negative position $-a_n$ and its gradient $-h_n = -\psi(a_n)$.

5.1 Initialization at the first plateau

Since $e_0 \le b_0/100$, we have $b_0 + 2e_0 < 1$. Equation (25) gives $g(1) = 1/2$ and hence $x_1 = b_0$. At that point $g(x_1) = e_0$, so

$$L_1 = \frac{1/2 - e_0}{1/2} = 1 - 2e_0 \in [99/100, 1).$$

The growth candidate is $\alpha_0 \sqrt{2/3 + \theta_0} = 1$, whereas the curvature candidate $1/\sqrt{2L_1^2 - 1}$ is strictly larger than one. Thus $\alpha_1 = 1$ and $\theta_1 = 1$, exactly the seed clock at $t_0 = 1$. The optional source calibration $\alpha_0 L_1 \in [1/\sqrt{2}, 2]$ is also satisfied.

5.2 Plateau motion, negative landing and return

Suppose stage n starts at $x_{t_n} = b_n$, with $\alpha_{t_n} = A_{n,0}$ and $\theta_{t_n} = v_{n,0}$. This has just been established for $n = 0$. For $0 \le j < N_n$, the points $b_n - e_n S_{n,j}$ lie in $[p_n, b_n]$. Their gradients all equal e_n. Every later secant within this plateau is zero, so the curvature candidate is $+\infty$ and induction gives

$$\begin{aligned} x_{t_n+j} &= b_n - e_n S_{n,j} && (0 \le j \le N_n), \\ \alpha_{t_n+j} &= A_{n,j}, \qquad \theta_{t_n+j} = v_{n,j} && (0 \le j < N_n). \end{aligned} \tag{26}$$

The last step leaves the plateau, because

$$e_n S_{n,N_n} = b_n + a_n, \qquad p_n + a_n = e_n A_{n,N_n-1}, \qquad x_{t_n+N_n} = -a_n. \tag{27}$$

Write $H = A_{n,N_n-1} = (p_n + a_n)/e_n$. The actual measured secant at this negative point is

$$L = \frac{h_n + e_n}{p_n + a_n}, \qquad HL = \frac{h_n + e_n}{e_n} > 1.$$

Its curvature candidate is therefore exactly

$$\frac{H}{\sqrt{2H^2L^2 - 1}} = \frac{p_n + a_n}{\sqrt{2(h_n + e_n)^2 - e_n^2}} = B_n, \qquad \frac{B_n}{H} = \delta_n < \frac{1}{4}. \tag{28}$$

The growth candidate has ratio $\sqrt{2/3 + v_{n,N_n-1}} > \sqrt{2/3} > 1/4$ to H. Thus the curvature candidate is strictly selected, and

$$\alpha_{t_n+N_n} = B_n, \qquad \theta_{t_n+N_n} = \delta_n, \qquad x_{t_n+N_n+1} = -a_n + B_n h_n = b_{n+1}. \tag{29}$$

In particular the seed negative landing is at time N_0+1, and its return is at time $N_0+2 = t_1$.

5.3 Entry to each subsequent plateau

At t_{n+1} the preceding displacement is $B_n h_n = a_n + b_{n+1}$. The two sampled gradients are $-h_n$ and e_{n+1}, so

$$B_n L_{t_{n+1}} = 1 + \frac{e_{n+1}}{h_n}.$$

The two candidate steps are

$$B_n \tau_{n+1}, \qquad \frac{B_n}{\sqrt{2(1 + e_{n+1}/h_n)^2 - 1}}. \tag{30}$$

The first is no larger if and only if

$$e_{n+1} \le h_n \left[\sqrt{\frac{1 + \tau_{n+1}^{-2}}{2}} - 1 \right] = \Gamma_{n+1}.$$

Our choice $e_{n+1} \le \Gamma_{n+1}/2 < \Gamma_{n+1}$ makes it strictly smaller. Therefore

$$\alpha_{t_{n+1}} = B_n \tau_{n+1} = A_{n+1,0}, \qquad \theta_{t_{n+1}} = \tau_{n+1} = v_{n+1,0}.$$

This closes the induction and proves (26)–(29) for every $n \ge 0$.

All sampled gradients are nonzero: the initial value is $1/2$, each positive plateau value is $e_n > 0$, and each negative value is $-h_n < 0$. Every selected step is finite and positive. Consequently

$$|x_k - x_{k-1}| = \alpha_{k-1}|g(x_{k-1})| > 0$$

at every positive index. The allowed infinity candidate on a plateau is never an undefined secant. The consecutive blocks $\{t_n, \dots, t_n + N_n\}$ cover every index $k \ge 1$, since $t_0 = 1$, each N_n is finite and $t_{n+1} \ge t_n + 3$.

Nor does any later interval alter past oracle values. The positive queries in stage n lie at or above $p_n > b_{n+1}$, and its connector preserves the value e_n at p_n. Later stages lie below b_{n+1}. The return to b_{n+1} uses only the fixed negative branch; the new value e_{n+1} is first queried at the next entry, where (30) applies. The entire function was defined before verifying these identities.

6 Convergence without a prescribed terminal rate

For $t_n \le k < t_{n+1}$, the actual formulas give $|x_k| \le b_n$. Since $b_n \le b_0 6^{-n} \to 0$ and the stage index tends to infinity with k, the whole sequence converges to zero. Continuity of g yields $g(x_k) \to 0$.

At $k_n = t_n + N_n$, where $n \ge 1$, the gradient magnitude is $h_n = \psi(cb_n)$. Equation (17) gives

$$k_n|g(x_{k_n})| \ge N_n h_n \ge n, \qquad \frac{|g(x_{k_n})|}{r_{k_n}} = \frac{h_n}{r_{t_n+N_n}} \ge n. \tag{31}$$

The indices k_n increase to infinity, proving both limsup statements in (3).

The summability follows directly, without invoking a convergence theorem. The initial term is $\alpha_0|g(x_0)|^2 = 1/4$. For every stage $n \ge 0$,

$$\begin{aligned} \sum_{k=t_n}^{t_n+N_n} \alpha_k |g(x_k)|^2 &= e_n^2 S_{n,N_n} + B_n h_n^2 \\ &= \tfrac{9}{8} b_n e_n + h_n(a_n + b_{n+1}) \\ &\le \left(\frac{9}{800} + \frac{7}{192}\right) b_n^2 = \frac{229}{4800} b_n^2. \end{aligned} \tag{32}$$

Here $h_n \le a_n = b_n/8$ and $b_{n+1} < b_n/6$ control the negative term. Thus

$$\sum_{k=0}^{\infty} \alpha_k |g(x_k)|^2 \le \frac{1}{4} + \frac{229}{4800} \sum_{n=0}^{\infty} b_n^2 < \infty.$$

This completes Theorem 1. The reset steps need not be bounded: indeed (12) and $b_{n+1} > b_n/10$ give

$$B_n = \frac{a_n + b_{n+1}}{h_n} > \frac{9}{5}\exp(64/b_n^2) \longrightarrow \infty.$$

Each is nevertheless finite at its own stage, and (32) controls its contribution.

Corollary 6.1. *For every positive sequence $s_k \to 0$, there is one fixed objective with the regularity, normalization and initialization of Theorem 1 whose actual trajectory converges to zero but satisfies*

$$\limsup_{k\to\infty} \frac{f(x_k) - f(0)}{s_k} = \infty.$$

Thus no common prescribed null rate for the last-objective gap holds over this class, even with instance-dependent multiplicative constants.

Proof. Apply the construction with $r_k = \sqrt{2s_k}$. For $a > 0$ write $h = \psi(a) \le a$. Since ψ is increasing and 1-Lipschitz,

$$\psi(u) \ge h - (a - u) \quad (a - h \le u \le a), \qquad f(-a) = \int_0^a \psi(u)\,du \ge \frac{h^2}{2}.$$

At the negative landings, (31) now gives

$$\frac{f(x_{k_n}) - f(0)}{s_{k_n}} \ge \left(\frac{|g(x_{k_n})|}{\sqrt{2s_{k_n}}}\right)^2 \ge n^2.$$

∎

The best-objective guarantee in [1] remains compatible with this corollary. Infinite differentiability here includes the minimum: all derivatives of g vanish at zero. It does not impose analyticity, strong convexity or a specified power-type gradient inequality. The theorem leaves rates under additional structural assumptions or different iterate selections untouched, and makes no assertion about general Kurdyka–Łojasiewicz properties.

7 Finite-order geometric terminal rates

This section proves Theorem 2. Its global convergence input is precisely the iterate-convergence conclusion of [1, Theorem 2]. The new conclusions here are permanent curvature-candidate inactivity, eventual sign preservation, the positive stepsize amplitude, and the exact terminal equivalents.

7.1 Well-definedness and the global convergence input

Convexity and uniqueness of the minimizer imply that g is nondecreasing, $g(0) = 0$, and

$$xg(x) > 0 \qquad (x \ne 0). \tag{33}$$

Indeed, a zero derivative of a differentiable convex function is equivalent to being a minimizer. Before termination, the growth candidate in (2) is finite and strictly positive, and the curvature candidate is either finite and strictly positive or $+\infty$. Thus every selected step is finite and strictly positive. Moreover

$$x_k - x_{k-1} = -\alpha_{k-1} g(x_{k-1}) \neq 0,$$

so each required secant is defined. The derivative is finite at finite points, and induction gives finite iterates and secants. This covers termination at $x_1 = 0$ as well as at any later zero iterate; no update is made after termination.

In the alternative infinite run, the hypotheses of [1, Theorem 2] hold: f is convex with locally Lipschitz gradient, and a solution exists. That theorem permits arbitrary x_0 and arbitrary $\alpha_0 > 0$. Its optional initial calibration

$$\alpha_0 L_1 \in [1/\sqrt{2}, 2]$$

is used for a particular best-objective constant, not for iterate convergence. The discussion preceding that theorem, Appendix B.1, and Remark 4 and the convergence proof in Appendix B.2 explicitly retain convergence without calibration. We import only the global iterate-convergence conclusion, which gives $x_k \to 0$ by uniqueness. All remaining assertions are proved below from the adaptive recurrence.

7.2 Identities for the actual adaptive minimum

For this section define the normalized step and successive derivative ratio by

$$z_k = \frac{\alpha_k g(x_k)}{x_k} > 0, \qquad R_{k+1} = \frac{g(x_{k+1})}{g(x_k)}.$$

The number q in (6) is the positive root of $q^2 = q + 2/3$, with $1 < q < 3/2$. The growth candidate and $\theta_0 = 1/3 < q$ imply by induction that

$$0 < \theta_k \leq q \qquad (k \geq 0). \tag{34}$$

The iterate update gives

$$\frac{x_{k+1}}{x_k} = 1 - z_k. \tag{35}$$

In an infinite nonzero run $z_k \neq 1$. Monotonicity of g and the direction of the gradient step give $R_{k+1} \leq 1$: for a positive derivative the step decreases x and cannot increase g; for a negative derivative the step increases x and cannot decrease g, with division by $g(x_k) < 0$ reversing the inequality. Hence the exact secant cancellation is

$$\alpha_k L_{k+1} = \frac{|g(x_{k+1}) - g(x_k)|}{|g(x_k)|} = 1 - R_{k+1}. \tag{36}$$

Consequently the original minimum is exactly

$$\theta_{k+1} = \min\left\{\sqrt{\tfrac{2}{3}+\theta_k},\ \frac{1}{\sqrt{[2(1-R_{k+1})^2-1]_+}}\right\}. \tag{37}$$

No eventual branch choice has been assumed.

For $x \neq 0$, put $H(x) = g(x)/(ax^p)$. Equation (33) and oddness of p give $H(x) > 0$, while (4) gives $H(x) \to 1$ at zero. Set $\varepsilon_k = H(x_{k+1})/H(x_k) \to 1$. Then

$$R_{k+1} = \varepsilon_k(1-z_k)^p, \qquad z_{k+1} = \theta_{k+1}\varepsilon_k z_k|1-z_k|^{p-1}. \tag{38}$$

These are derivative-value identities. Neither a derivative of H nor an estimate on f'' is used. Fix a finite N such that

$$\frac{10}{11} \le \varepsilon_k \le \frac{11}{10} \qquad (k \ge N). \tag{39}$$

7.3 Escape from overshoots and a permanent interval

Suppose $k \ge N$ and $z_k > 1$. Put $s = z_k - 1 > 0$ and $B = \varepsilon_k s^p > 0$. Then $R_{k+1} = -B$, so the actual curvature candidate in (37) gives

$$\theta_{k+1} \le \frac{1}{\sqrt{1+4B+2B^2}}, \qquad z_{k+1} \le \frac{(1+s)B}{s\sqrt{1+4B+2B^2}}.$$

For $0 < s \le 1$, $B \le (11/10)s^p \le (11/10)s$. Using $\sqrt{1+4B+2B^2} > 1+B$ and monotonicity of $B/(1+B)$,

$$z_{k+1} < \frac{(1+s)B}{s(1+B)} \le \frac{(11/10)(1+s)}{1+(11/10)s} \le \frac{11}{10} < \frac{3}{2}.$$

For $s \ge 1$, the inequality $\sqrt{1+4B+2B^2} > \sqrt{2}B$ instead yields

$$z_{k+1} < \frac{1+1/s}{\sqrt{2}} \le \sqrt{2} < \frac{3}{2}.$$

Every sufficiently late overshoot therefore gives $0 < z_{k+1} < 3/2$. If this value exceeds one, equations (34), (38) and (39), with $p-1 \ge 2$, give

$$z_{k+2} \le q\frac{11}{10}z_{k+1}(z_{k+1}-1)^{p-1} < \frac{3}{2}\frac{11}{10}\frac{3}{2}\left(\frac{1}{2}\right)^2 = \frac{99}{160} < 1. \tag{40}$$

Equality to one would terminate the run and is excluded in the infinite alternative. It follows, whether or not $z_N > 1$, that some $j \in \{N, N+1, N+2\}$ has $0 < z_j < 1$.

For $0 < z < 1$, differentiation gives

$$\max_{0\le z\le 1} z(1-z)^{p-1} = \frac{1}{p}\left(1-\frac{1}{p}\right)^{p-1}.$$

For real $p \geq 3$, $(1-1/p)^{p-1} \leq 4/9$: the logarithmic derivative with respect to p is $\log(1-1/p) + 1/p < 0$, and the value at $p = 3$ is $4/9$. Thus for $k \geq N$ with $0 < z_k < 1$,

$$0 < z_{k+1} \leq \frac{11q}{10p}\left(1 - \frac{1}{p}\right)^{p-1} < \frac{11}{15p} < b := \frac{3}{4p} < 1. \tag{41}$$

Induction, starting at j, proves $0 < z_k < b$ for every $k \geq j+1$. By (35), the sign of x_k is constant from index j onward. Denote this sign by $\sigma \in \{-1, 1\}$.

7.4 Strict permanent inactivity of the curvature candidate

For $k \geq j+1$, the preceding bounds give

$$R_{k+1} \geq \frac{10}{11}\left(1 - \frac{3}{4p}\right)^{p} \geq \frac{10}{11}\left(\frac{3}{4}\right)^{3} = \frac{135}{352} > \frac{3}{8}. \tag{42}$$

To justify the second inequality, for fixed $d = 3/4$ the function $(1-d/p)^p$ increases for $p > d$. Its logarithmic derivative is

$$\log(1 - d/p) + \frac{d/p}{1 - d/p} > 0;$$

the expression $\log(1-y) + y/(1-y)$ is zero at $y = 0$ and has positive derivative for $0 < y < 1$. Combining (42) with $R_{k+1} \leq 1$ and (36) yields

$$0 \leq \alpha_k L_{k+1} < \frac{5}{8}, \qquad 2\alpha_k^2 L_{k+1}^2 < \frac{25}{32} < 1. \tag{43}$$

The curvature-candidate denominator at index $k+1$ is therefore exactly zero, with a positive numerator. The candidate is $+\infty$ for every such index, and the actual rule reduces to

$$\alpha_{k+1} = \alpha_k \sqrt{\tfrac{2}{3} + \theta_k}, \qquad \theta_{k+1} = \sqrt{\tfrac{2}{3} + \theta_k}. \tag{44}$$

Taking $K = j+2$ proves (5). This strict, permanent conclusion follows from the actual minimum, including its control of the preceding sign-changing steps.

7.5 The positive amplitude with the original index

The map $\Phi(u) = \sqrt{2/3 + u}$ has fixed point q and is Lipschitz on $[0, q]$ with constant

$$\lambda = \frac{1}{2\sqrt{2/3}} = \sqrt{\frac{3}{8}} < 1.$$

Equation (44) gives $|\theta_k - q| \leq \lambda |\theta_{k-1} - q|$ for $k \geq K$, and hence

$$\theta_k \to q, \qquad \sum_{k=K}^{\infty} |\theta_k/q - 1| < \infty.$$

Every factor θ_k/q is positive. For factors sufficiently close to one, $|\log(\theta_k/q)| \le 2|\theta_k/q - 1|$, while the finitely many earlier logarithms are finite. The product therefore converges to a finite strictly positive limit. With no reindexing,

$$\frac{\alpha_k}{q^k} = \frac{\alpha_{K-1}}{q^{K-1}} \prod_{i=K}^{k} \frac{\theta_i}{q} \longrightarrow A \in (0, \infty). \tag{45}$$

No rate for $H(x_k) - 1$ is needed for this conclusion.

7.6 Convergence of the normalized step

On $[0, b]$, define $F(z) = qz(1 - z)^{p-1}$. By (38),

$$z_{k+1} = v_k F(z_k), \qquad v_k = (\theta_{k+1}/q)\varepsilon_k \longrightarrow 1. \tag{46}$$

Since $pb = 3/4$, we have $F'(z) = q(1 - z)^{p-2}(1 - pz) > 0$ on this interval. Also $F(z)/z = q(1 - z)^{p-1}$ decreases strictly for positive z, starts above one at zero, and

$$F(b) \le \frac{q}{p}\left(1 - \frac{1}{p}\right)^{p-1} < \frac{2}{3p} < b.$$

Thus its unique positive fixed point in $[0, b]$ is

$$\zeta_p = 1 - q^{-1/(p-1)} \in (0, b). \tag{47}$$

A positive lower bound must still be proved. Choose $v_- \in (1/q, 1)$ and a sufficiently late $M \ge j + 1$ so that $v_k \ge v_-$ for all $k \ge M$. The increasing map $F_- = v_- F$ has positive fixed point

$$\zeta_- = 1 - (v_- q)^{-1/(p-1)} \in (0, \zeta_p).$$

Set $d = \min\{z_M, \zeta_-\} > 0$. For $0 < z \le \zeta_-$ we have $F_-(z) \ge z$. Since F_- increases on $[0, b]$, equation (46) gives by induction

$$d \le z_k < b \qquad (k \ge M). \tag{48}$$

Let $\ell = \liminf z_k$ and $u = \limsup z_k$. Continuity and monotonicity of F on $[d, b]$ imply $\liminf F(z_k) = F(\ell)$ and $\limsup F(z_k) = F(u)$. The difference $v_k F(z_k) - F(z_k)$ tends to zero, and shifting the sequence preserves both limits. Thus

$$\ell = F(\ell), \qquad u = F(u).$$

Both are positive by (48), so uniqueness of the positive fixed point gives $z_k \to \zeta_p$. Along with (45), this proves (7).

7.7 Iterate, gradient and objective equivalents

Since $p-1$ is even, $z_k = \alpha_k a|x_k|^{p-1}H(x_k)$. Therefore

$$q^k|x_k|^{p-1} = \frac{z_k}{a(\alpha_k/q^k)H(x_k)} \longrightarrow \frac{\zeta_p}{aA}.$$

The eventual sign then proves $q^{k/(p-1)}x_k \to \sigma D$, with the stated $D > 0$. Next,

$$q^{pk/(p-1)}g(x_k) = aH(x_k)\big(q^{k/(p-1)}x_k\big)^p \longrightarrow \sigma a D^p.$$

Finally, integration of (4) gives

$$f(x) - f(0) = \frac{a}{p+1}|x|^{p+1}(1+o(1)). \tag{49}$$

Indeed, bound H between $1-\eta$ and $1+\eta$ on the interval between zero and x, then integrate $ax^pH(x)$ with the appropriate orientation. This works on both sides of zero and needs no second derivative. Substituting the iterate equivalent into (49) proves the last limit in (8). All amplitudes are finite and strictly positive in absolute value. Quotients of consecutive equivalents give (9); their denominators are nonzero because the trajectory is infinite and the minimizer is unique. This completes Theorem 2.

8 Analytic minima and scope

Corollary 8.1. *Let a finite scalar convex real-analytic objective on $\mathbb{R}$ have an isolated minimum at zero, with first nonzero Taylor term of order $2m \geq 4$. Then the original AdGD2 rule, with any $x_0 \neq 0$, any $\alpha_0 > 0$ and $\theta_0 = 1/3$, either terminates at zero or has all conclusions of Theorem 2, with $p = 2m-1$ and the positive leading derivative coefficient a.*

Proof. Write $f(x) = f(0) + c_m x^{2m} + O(x^{2m+1})$ with $c_m > 0$. Convexity makes an isolated minimizer unique. Real analyticity gives a locally Lipschitz derivative, and differentiating the convergent Taylor expansion yields

$$g(x) = 2mc_m x^{2m-1} + O(x^{2m}).$$

Theorem 2 applies with $p = 2m-1 \geq 3$ and $a = 2mc_m > 0$. A minimum at another point is handled by translation. ∎

The all-orders flat minimum of Theorem 1 lies outside (4). In particular, (49) has a nonzero finite-order leading term, whereas the constructed slow objective has all positive-order derivatives zero at its minimizer. Smoothness alone therefore leaves room for arbitrarily slow terminal behavior, while the specific finite-order derivative-value hypothesis forces geometric terminal equivalents for this scalar rule.

The slow result is an obstruction with quantifiers $\forall r\ \exists f_r$ and a subsequence witness. The finite-order result is an instancewise asymptotic theorem with an instance-dependent amplitude and entry time. Neither yields a uniform finite-horizon complexity statement, a theorem for $p = 1$, or a multidimensional extension. Both preserve the convergence and best-iterate conclusions of the source algorithm. The smooth construction and the finite-order analysis are two conclusions of this note; the global convergence theorem imported in the latter remains attributable to [1].

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